

Algorithms

Lecture #17

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Dynamic Programming

Dynamic Programming

Fibonacci Number Iterative Algorithm

Fibonacci Number: Iterative Algorithm

This gives us our first explicit dynamic programming algorithm.

ITERFIB(n)

```
1    $F[0] \leftarrow 0$ 
2    $F[1] \leftarrow 1$ 
3   for  $i \leftarrow 2$  to  $n$ 
4   do
5        $F[i] \leftarrow F[i - 1] + F[i - 2]$ 
6   return  $F[n]$ 
```

Fibonacci Number Iterative Algorithm

Fibonacci Number: Iterative Algorithm

- This algorithm clearly takes only $O(n)$ time to compute F_n .
- By contrast, the original recursive algorithm takes $\Theta(\Phi^n)$, $\Phi = \frac{1+\sqrt{5}}{2} \approx 1.618$.
- ITERFIB achieves an exponential speedup over the original recursive algorithm.

Dynamic Programming

Dynamic Programming

- Dynamic programming is essentially recursion without repetition.
- Developing a dynamic programming algorithm generally involves two separate steps:

Dynamic Programming

Dynamic Programming

- **Formulate problem recursively.** Write down a formula for the whole problem as a simple combination of answers to smaller subproblems.
- **Build solution to recurrence from bottom up.** Write an algorithm that starts with base cases and works its way up to the final solution.

Dynamic Programming

Dynamic Programming

- Dynamic programming algorithms need to store the results of intermediate subproblems.
- This is often *but not always* done with some kind of table.

Edit Distance

Edit Distance

- Introduced by Levenshtein in 1966.
- Definition: Minimum number of edit operations to transform one string to another
- Possible edit operations
 - Symbol insertion (I)
 - Symbol deletion (D)
 - Symbol substitution (S)

Edit Distance

Edit Distance

- For example, the edit distance between FOOD and MONEY is at most four:

FOOD → MOOD → MON_^D
→ MONED → MONEY

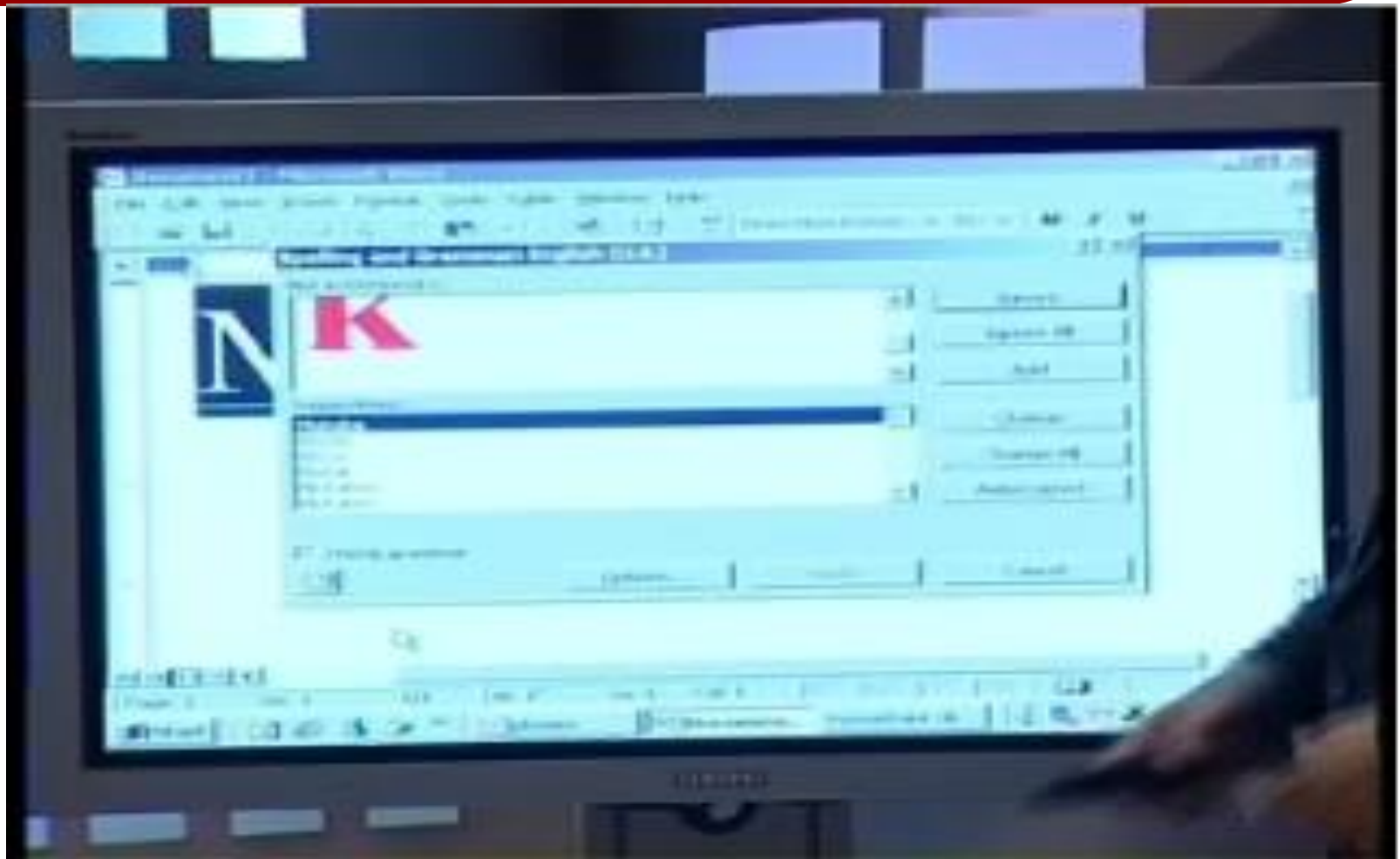
Edit Distance: Application

Edit Distance: Applications

Spelling Correction

If a text contains a word that is not in the dictionary, a 'close' word, i.e. one with a small edit distance, may be suggested as a correction.

Mestak



Edit Distance: Application

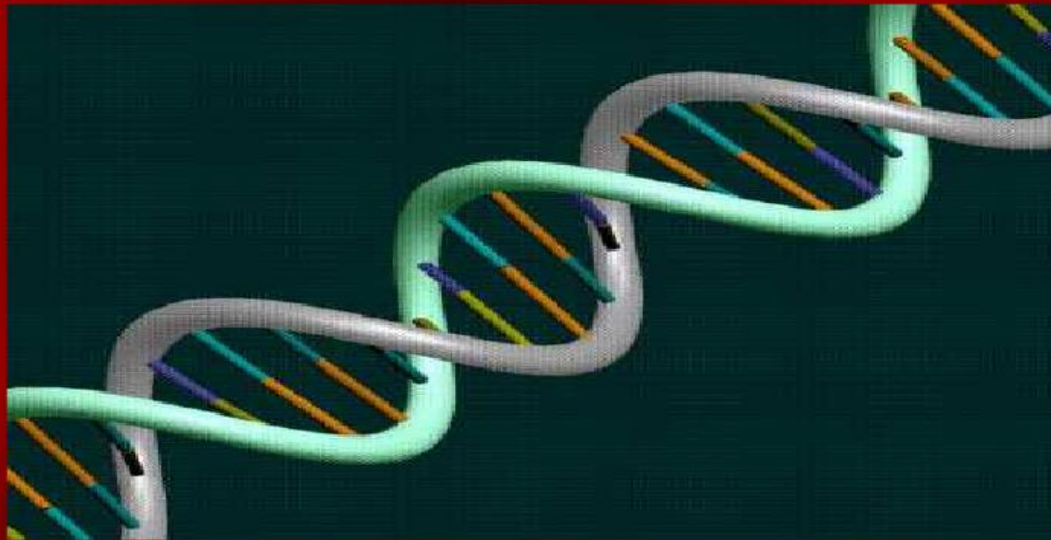
Edit Distance: Applications

Plagiarism Detection

The edit distance provides an indication of similarity that might be too close in some situations.

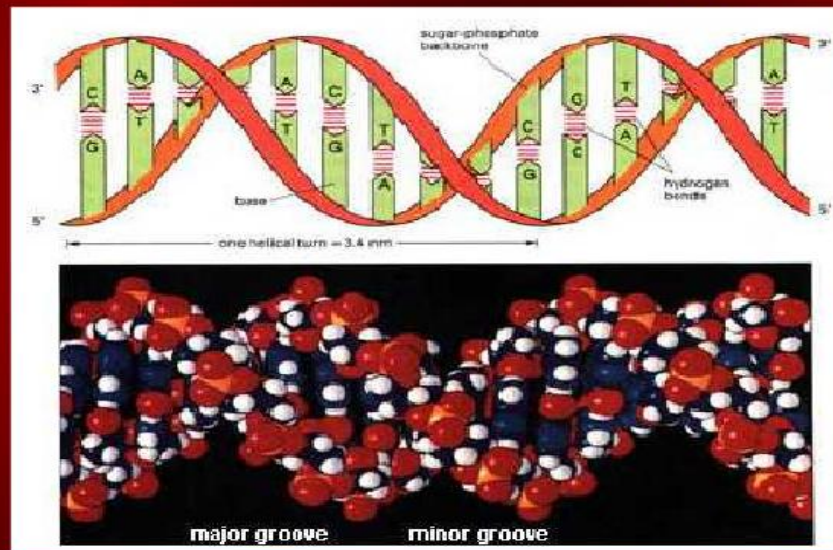
Computational Molecular Biology

Computational Molecular Biology

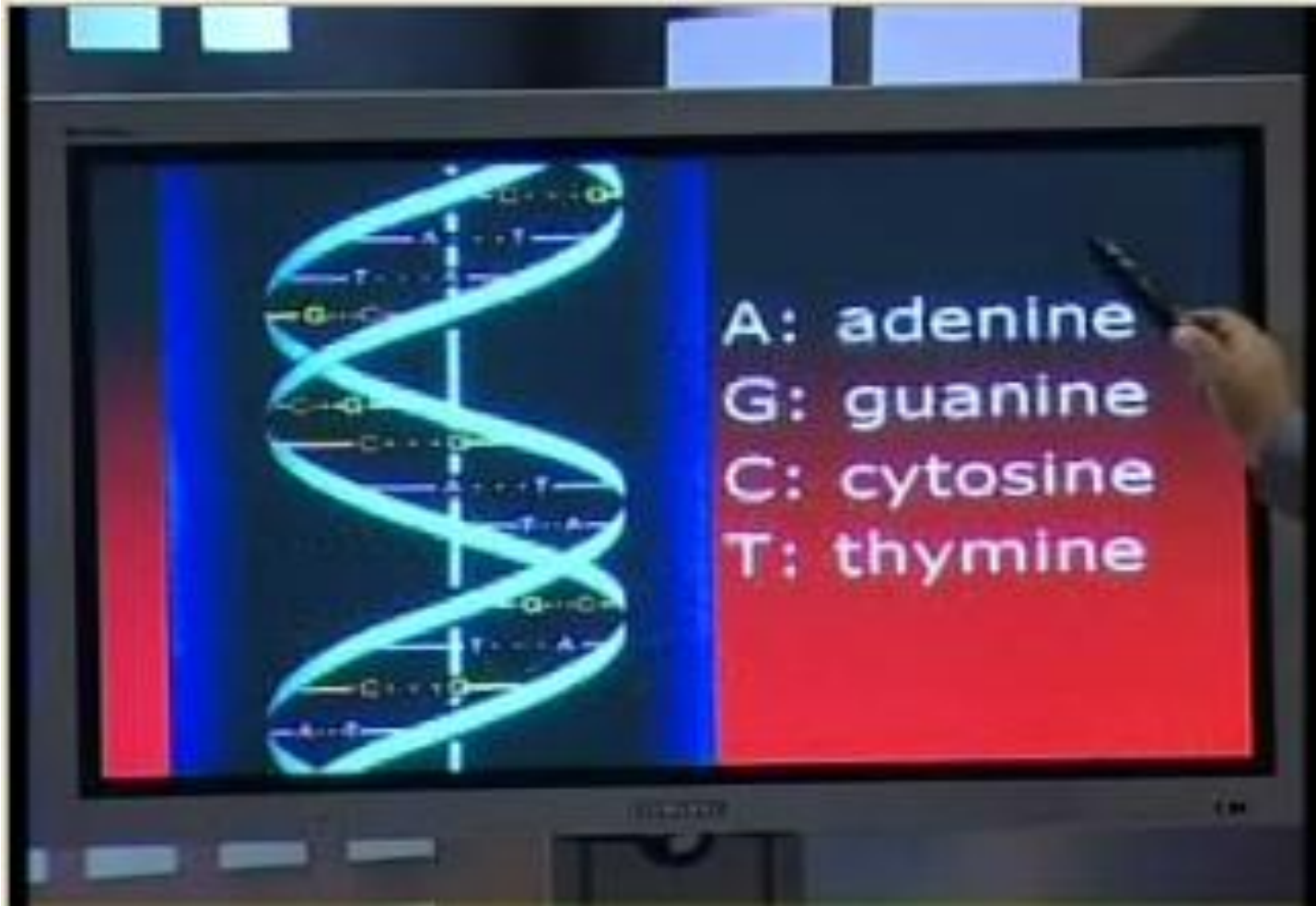


Computational Molecular Biology

Computational Molecular Biology



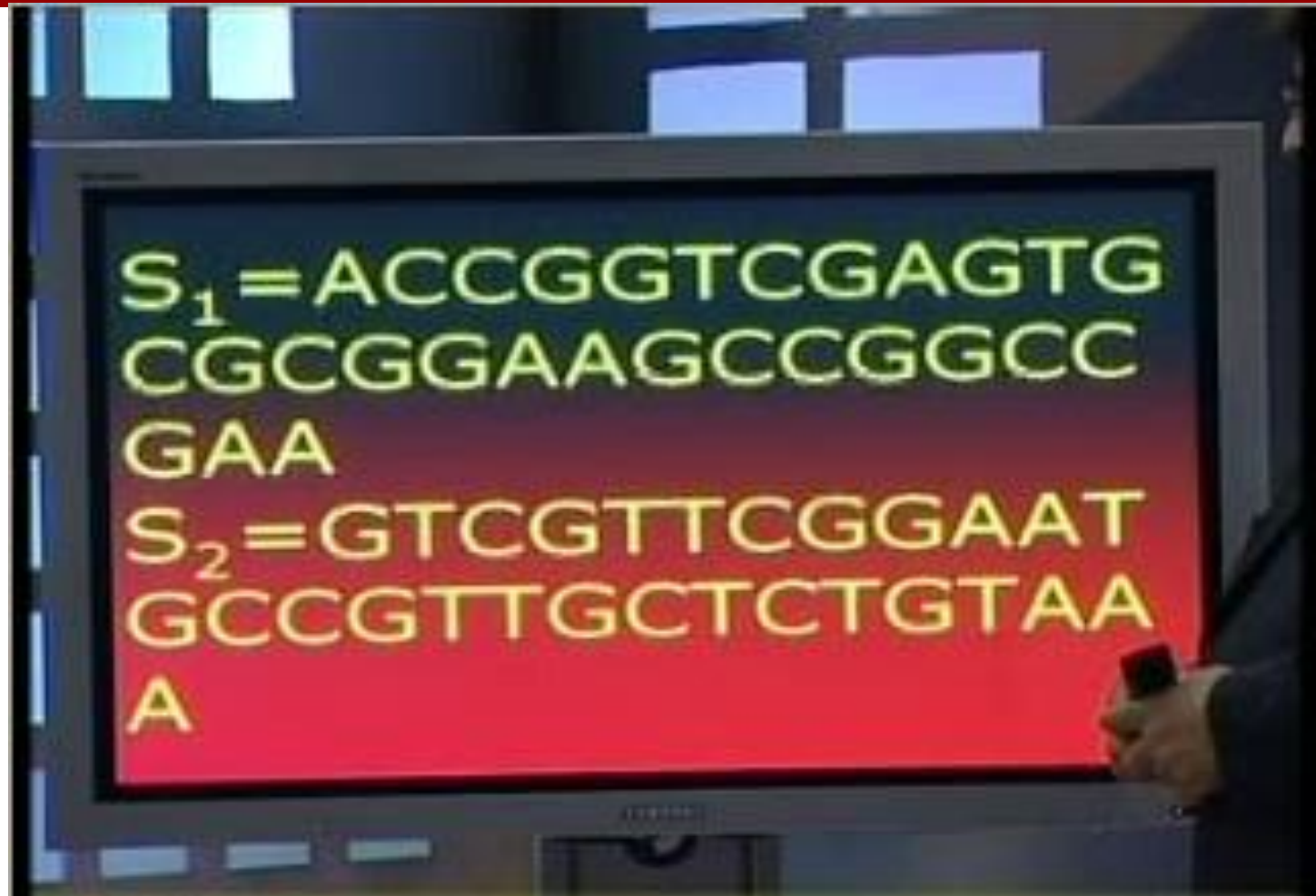
- A-adenine
- G-guanine
- C-cytosine
- T-thymine







Longest Common Subsequence (LCS)



Human beta globin region on
chromosome 11 (73321)

```
gaattotaatctccctctcaaccctaca  
gtcaccatttggtatatataaagatgtg  
ttgtctactgtctagtatccctcaagta  
gtgtcaggaattagtcatttaaataagto  
tgcaagccaggagtgggtggctcatgtct  
gtaattccagcactggagagggtagaagt  
gggaggactgottgagotcaagagtttg  
atattatcctggacaacatagcaagacc
```

LCS

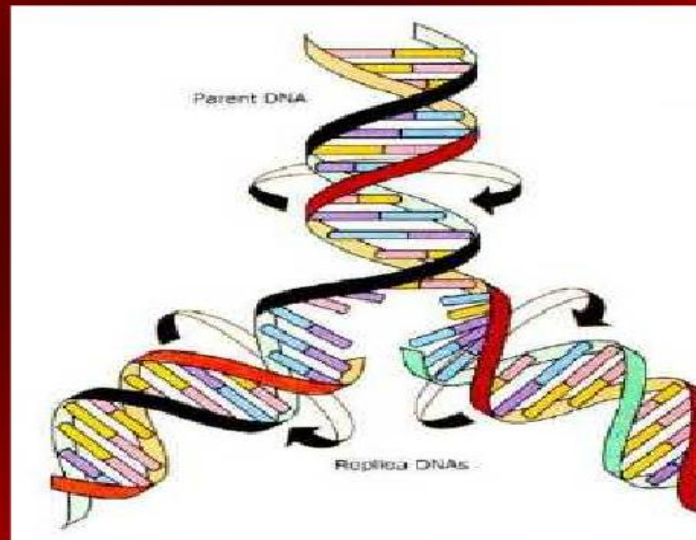
$S_3 =$ GTCGTCGGAAGC
CGGCCGAA

A hand is visible on the right side of the screen, pointing towards the text. The hand is holding a pen or pointer, and the index finger is extended towards the text.



Computational Molecular Biology

Computational Molecular Biology



Edit Distance: Application

Edit Distance: Applications

Computational Molecular Biology

DNA Replication errors lead to nucleotide:

- Substitutions
- Insertions
- Deletion.

Edit Distance: Application

Edit Distance: Applications

Human beta globin region on chromosome 11

1	gaattcta	ctccctca	accctacgt	caccatttg	gtatattaa	gatgtgtgt
61	ctactgtcta	gtatccctca	agtagtgtca	ggaattagtc	atttaaata	tctgcaagcc
121	aggagtgg	gctcatgtct	gtaattccag	cactggagag	gtagaagtgg	gaggactgct
181	tgagctcaag	agtttgatat	tatcctggac	aacatagcaa	gacctcgtct	ctacttaaaa
241	aaaaaaaaat	tagccaggca	tgtgatgtac	acctgtagtc	ccagctactc	aggaggccga
301	aatgggagga	tcccttgagc	tcaggaggtc	aaggctgcag	tgagacatga	tcttgccact
361	gcactccagc	ctggacagca	gagtgaac	ttgcctcacg	aaacagaata	caaaaacaaa
421	caaacaaaa	actgctccgc	aatgcgcttc	cttgatgctc	taccacatag	gtctgggtac
481	tttgtagaca	ttatctcatt	gctgttcgta	attgtagat	taattttgta	atattgatat
541	tattcctaga	aagctgaggc	ctcaagatga	taacttttat	tttctggact	tgtaatagct
601	ttctcttgta	ttcaccatgt	tgtaactttc	ttagagtagt	aacaatataa	agttattgtg
661	agtttttgca	aacacagcaa	acacaacgac	ccatatagac	attgatgtga	aattgtctat
721	tgtcaattta	tgggaaaaca	agtatgtact	tttctacta	agccattgaa	acaggaataa
781	cagaacaaga	ttgaaagaat	acattttccg	aaattacttg	agtattatac	aaagacaagc

Edit Distance: Application

Edit Distance: Applications

841	acgtggacct	gggaggaggg	ttattgtcca	tgactgggtgt	gtggagacaa	atgcaggttt
901	ataatagatg	ggatggcatc	tagcgcaatg	actttgcat	cacttttaga	gagctcttg
961	ggaccccagt	acacaagagg	ggacgcaggg	tatatgtaga	catctcattc	ttttcttag
1021	tgtgagaata	agaatagcca	tgacctgagt	ttatagacaa	tgagcccttt	tctctctccc
1081	actcagcagc	tatgagatgg	cttgccctgc	ctctctacta	ggctgactca	ctccaaggcc
1141	cagcaatggg	cagggctctg	tcagggcttt	gatagcacta	tctgcagagc	cagggccgag
1201	aaggggtgga	ctccagagac	tctccctccc	attcccagac	agggtttgct	tatttatgca
1261	tttaaatgat	atatttattt	taaaagaaat	aacaggagac	tgcccagccc	tggctgtgac
1321	atggaaacta	tgtagaatat	tttgggttcc	atTTTTTTT	ccttctttca	gttagaggaa
1381	aaggggctca	ctgcacatac	actagacaga	aagtcaggag	ctttgaatcc	aagcctgac
1441	atttccatgt	catactgaga	aagtcccccac	ccttctctga	gcctcagttt	ctctttttat
1501	aagtaggagt	ctggagtaaa	tgatttccaa	tggctctcat	ttcaatacaa	aatttccgtt
1561	tattaaatgc	atgagcttct	gttactccaa	gactgagaag	gaaattgaac	ctgagactca
1621	ttgactggca	agatgtcccc	agaggctctc	attcagcaat	aaaattctca	ccttcaccca
1681	ggcccactga	gtgtcagatt	tgcatgcact	agttcacgtg	tgtaaaaagg	aggatgcttc

Edit Distance: Application

Edit Distance: Applications

1741	tttcctttgt	attctcacat	acctttagga	aagaacttag	cacccttccc	acacagccat
1801	cccaataact	catttcagtg	actcaaccct	tgactttata	aaagtcttgg	gcagtataga
1861	gcagagatta	agagtacaga	tgctggagcc	agaccacctg	agtgattagt	gactcagttt
1921	ctcttagtaa	ttgtatgact	cagtttcttc	atctgtaaaa	tggagggttt	tttaattagt
1981	ttgttttga	gaaagggctc	cactctgtca	cccaaagggg	agtgtagtgg	caaaatctcg
2041	gctcactgca	acttgcactt	cccaggctca	agcggctctc	ccacctcaac	atcctgagta
2101	gctggaacca	caggtacaca	ccaccatacc	tcgctaattt	tttgatattt	tggtagagat
2161	ggggtttcac	atgttacaca	ggatgggtctc	agactccgga	gctcaagcaa	tctgcccacc
2221	tcagccttcc	aaagtgtctg	gattataagc	atgattacag	gagttttaac	aggctcataa
2281	gattgttctg	cagcccagtg	gagttaatac	atgcaaagag	tttaaagcag	tgacttataa
2341	atgctaacta	ctctagaaat	gtttgctagt	atttttgtt	taactgcaat	cattcttgct
2401	gcaggtgaaa	actagtgttc	tgtactttat	gccattcat	ctttaactgt	aataataaaa
2461	ataactgaca	tttattgaag	gctatcagag	actgtaatta	gtgctttgca	taattaatca
2521	tatttaatac	tcttggttc	tttcaggtag	atactattat	tatccccatt	ttactacagt
2581	taaaaaaact	acctctcaac	ttgtcaagc	atacactctc	acacacacaa	acataaacta
2641	ctagcaaata	gtagaattga	gatttggctc	taattatgtc	tttgctcact	atccaataaa
2701	tatttattga	catgtacttc	ttggcagtct	gtatgctgga	tgctggggat	acaaagatgt

Edit Distance: Application

Edit Distance: Applications

72841	ggattaagaa	aatgtggcac	atatacacca	tggaatacta	tcgagccata	aaaaatgatg
72901	agttcatgtc	ctttgtaggg	acatggatga	agctggaaac	tatcattctc	agcaaactat
72961	cacaaggaca	ataaaccaaa	caccgcatgt	tctcactcat	agggtgggaat	tgaacaatga
73021	gaacacatgg	acacatgaag	aggaacatca	cactctgggg	actgttatgg	ggtggggggc
73081	aggggcaggg	atagcactag	gagatatacc	taatgctaaa	tgacgagtta	atgggtgcag
73141	cacaccaaca	tggcacatgt	atacatatat	aacaaacctg	ccgttgtgca	catgtaccct
73201	aaaacttgaa	gtataataat	aaaaaaaagt	tatcctatta	aaactgatct	cacacatccg
73261	tagagccatt	atcaagtctt	tctctttgaa	acagacagaa	atttagtggt	ttctcagtca
73321	gttaac					

Edit Distance: Application

Edit Distance: Applications

Computational Molecular Biology

Similarities in DNA Sequences can provide

- Clue to common evolutionary origin
- Clue to common function

Edit Distance: Application

Edit Distance: Applications

Speech Recognition

- Algorithms similar to those for the edit-distance problem are used in some speech recognition systems.
- Find a close match between a new utterance and one in a library of classified utterances.

Edit Distance

Edit Distance

A better way to display this editing process is to place the words above the other:

M	A	_	T	H	S
A	_	R	T	_	S

Edit Distance

Edit Distance

<i>S</i>	<i>D</i>	<i>I</i>	<i>M</i>	<i>D</i>	<i>M</i>
M	A	_	T	H	S
A	_	R	T	_	S

- The first word has a gap for every insertion (I) and the second word has a gap for every deletion (D).
- Columns with two different characters correspond to substitutions (S).
- Matches (M) do not Count

Edit Distance

Edit Distance

<i>S</i>	<i>D</i>	<i>I</i>	<i>M</i>	<i>D</i>	<i>M</i>
M	A	_	T	H	S
A	_	R	T	_	S

- **Edit transcript:** A string over the alphabet M, S, I, D that describes a transformation of one string into another Example.

<i>S</i>	<i>D</i>	<i>I</i>	<i>M</i>	<i>D</i>	<i>M</i>
1+	1+	1+	0+	1+	0+ = 4

Edit Distance

Edit Distance

DP

Formulation

Edit Distance